

CY 303 : Fundamentals of Inorganic Chemistry

Course Code : CY 303

Course Name : Fundamentals of Inorganic Chemistry

L-T-P-C : 3-0-0-3

Intended for : BS Chemical Sciences

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Periodic properties:** Atomic Structure, electronic configuration, Chemical periodicity and periodic anomalies, Size of atoms and ions, Effective nuclear charge, Screening effect, Ionization energy, Electronegativity, Electron affinity, Lanthanide contraction, Fajan's rules. (8 Hours)
- **Concepts of acids and bases:** Theories of acids and bases, Bronsted and Lewis acids and bases, Gas phase versus solution acidity, leveling effects of solvents, Concepts of pH, pKa, pKb, Hardness and softness, surface acidity. (8 Hours)
- **Principles of electrochemistry:** Oxidation and reduction, Redox potential and stability, Electrode potentials, Nernst equation, Frost, Latimer and Pourbaix diagrams. (6 Hours)
- **s & p block elements:** Structure and properties of s and p block elements, and their compounds like hydrides, oxides, and halides, biological functions of inorganic elements in organisms. (8 Hours)
- **Transition elements:** Coordination complexes, Isomerism, Theories of metal-ligand bonding and their limitations, Valence bond theory, Spectrochemical series of ligands, Crystal field theory, Splitting of d orbitals in octahedral, tetrahedral and square planar complexes, Low-spin and high-spin complexes, Brief introduction to color and magnetism. (8 Hours)
- **Introduction to nuclear chemistry:** Nuclear reactions, fission and fusion, radio analytical techniques. (4 Hours)

Textbooks:

1. Ajai Kumar, **Basic Inorganic Chemistry**, 2nd Edition, Aaryush Education, 2019.
2. J. E. Huheey, E. A. Keiter and R. L. Keiter, **Inorganic Chemistry: Principles of Structure and Reactivity**, 4th edition, Pearson Education Inc., 2000.

References:

1. . F. A. Cotton, G. Wilkinson, C. A. Murillo and M. Bochmann, **Advanced Inorganic Chemistry**, 6th edition, Wiley, 2007.
2. B. Douglas, D. McDaniel and J. Alexander, **Concepts and Models of Inorganic Chemistry**, 3rd edition, Wiley, 2006.
3. J. D. Lee, **Concise Inorganic Chemistry**, 5th edition, Wiley, 2010.



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4. P. Atkins et al, **Shriver & Atkins' Inorganic Chemistry**, 5th edition, W. H. Freeman and Company, 2010.